

INVENTORY MANAGEMENT SYSTEM

ABSTRACT:

The Inventory Management System (IMS) using Linked List is a data-structure-driven application designed to efficiently manage and manipulate inventory records through the dynamic memory capabilities of linked lists. Unlike array-based systems that suffer from fixed-size limitations, this system leverages the flexibility of linked lists to handle inventory data that grows or shrinks in real time without memory wastage.

Each node in the linked list represents a unique inventory item, encapsulating attributes such as Item ID, Item Name, Category, Quantity, Price, and Supplier Details. The linked structure allows seamless insertion, deletion, traversal, and searching of inventory records at any position — making operations like adding new stock, removing discontinued items, or updating quantities highly efficient.

Core operations implemented include: Insert at beginning / end / position — for adding new inventory items Delete by Item ID — for removing sold-out or discontinued products Search / Traversal — for locating specific items or displaying full inventory Sort (by price, quantity, or ID) — using linked list sorting algorithms like Merge Sort Update — for modifying stock levels or item details The system is particularly suitable as an academic or foundational project demonstrating how fundamental data structures can solve real-world inventory challenges. It highlights the advantages of dynamic memory allocation over static structures and serves as a stepping stone toward more complex database-backed inventory solutions.

Keywords: Linked List, Inventory Management, Dynamic Memory Allocation, Node, Insertion, Deletion, Traversal, Data Structures.

INTRODUCTION:

In today's business world, managing inventory efficiently is essential for smooth operations. Manual methods of tracking stock are slow, error-prone,

and inefficient. An Inventory Management System (IMS) solves this by automating the process of adding, updating, searching, and deleting inventory records.

This project implements an IMS using a Linked List data structure, where each node stores item details such as ID, Name, Quantity, Price, and Supplier. Linked lists offer dynamic memory allocation, making it easy to insert and delete records without memory wastage. The goal of this project is to demonstrate how data structures can be applied to real-world problems in an efficient and organized way.

ALGORITHM:

Step 1: Start

Step 2: Define Node Structure

Each node represents one item:

Node:

Item_Id

Item_Name

quantity

price

next (pointer to next node)

Step 3: Initialize

Create a pointer: head = NULL

Step 4: Display Menu (Repeat until exit)

Add Item

Update Item

Delete Item

Search Item

Display All Items

Exit

Step 5: Perform Operations

5a. Add Item

Create new node newNode

Input item details

Set newNode->next = NULL

If head == NULL

head = newNode

Else

Traverse to last node

Attach new node at end

5b . Update Item

Input item_id

Set temp = head

Traverse list:

If temp->item_id == given_id

Update quantity/price

Stop If not found -> print "Item not found"

5c . Delete Item

Input item_id

Set:

temp = head

prev = NULL

Traverse list:

If item found:

If deleting first node:

head = temp->next

Else:

prev->next = temp->next

Free temp

Stop

If not found -> print "Item not found"

5d . Search Item

Input item_id

Set temp = head

Traverse:

If found -> display details

If not found -> print "Item not found"

5e . Display All Items

Set temp = head

If head == NULL

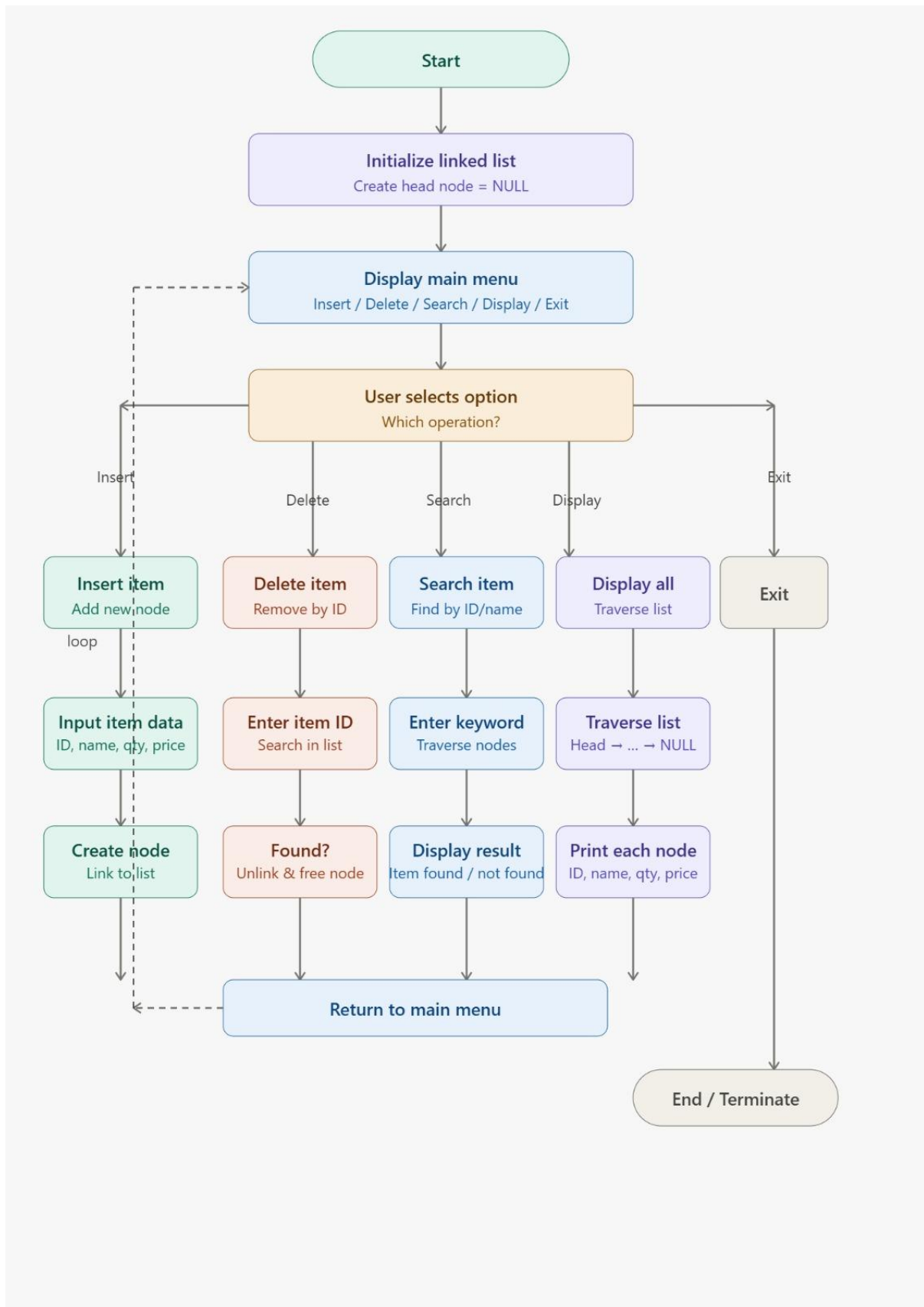
Print "Inventory empty"

Else Traverse list Print each node details

5f . Exit Stop program

Step 6:Stop

FLOW CHART:

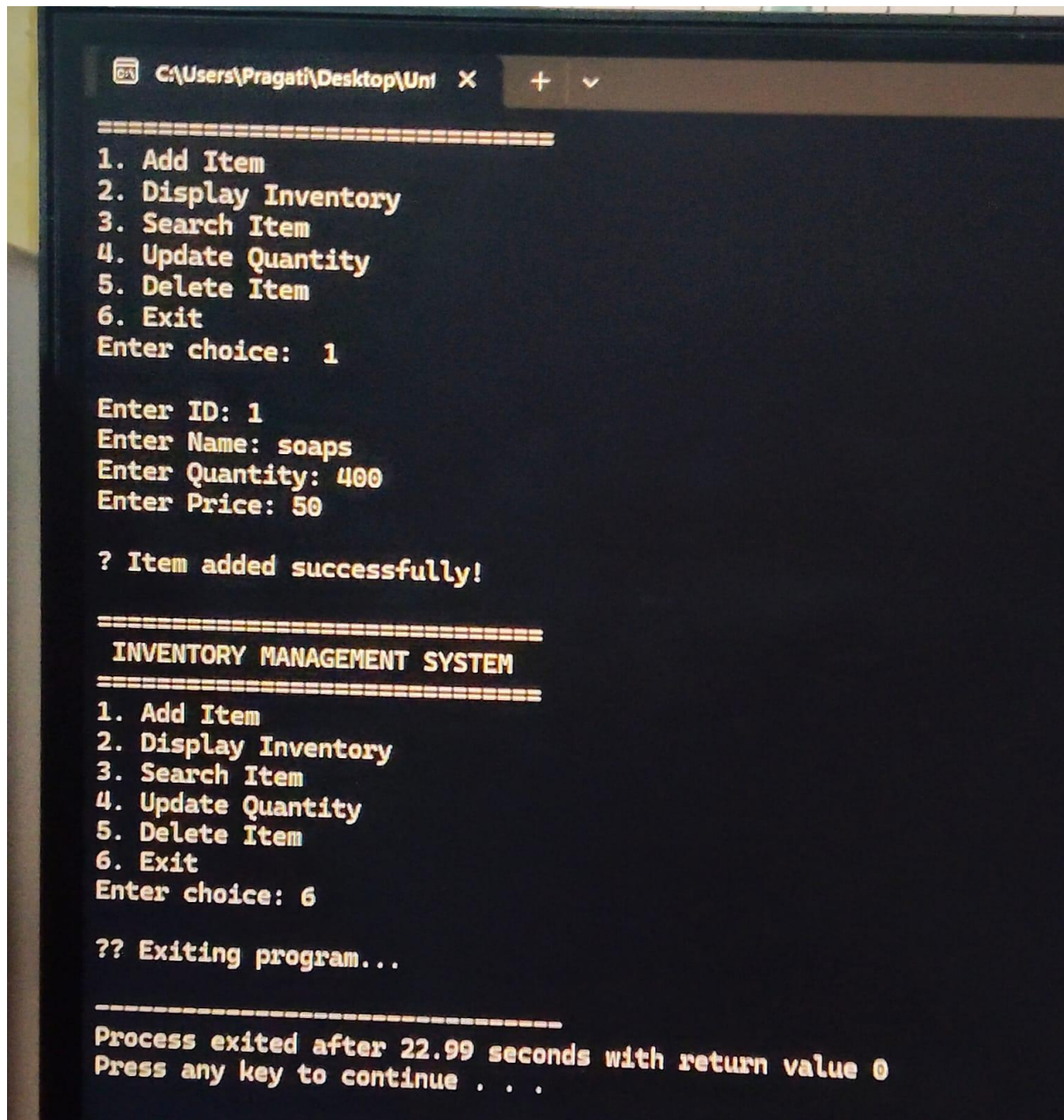


METHODOLOGY:

The Inventory Management System (IMS) project is designed to efficiently manage and control stock within an organization. The core ideology of this system is to ensure accuracy, automation, and real-time tracking of inventory data. The system follows a centralized approach where all inventory information is stored in one place, making it easy to access, update, and manage. It reduces manual work by automating key processes such as stock updates, billing, and report generation. Another important aspect of the ideology is maintaining a balance between supply and demand.

The system helps avoid overstocking and understocking, ensuring that the right quantity of products is available at the right time. The project also focuses on data-driven decision-making by generating useful reports such as stock status, sales trends, and inventory usage. These insights help in planning and improving business operations. Security and reliability are also key principles, with user authentication and controlled access to ensure data safety. Overall, the ideology of the Inventory Management System is to create a simple, efficient, and scalable solution that minimizes errors, saves time, reduces cost.

RESULT:



```
C:\Users\Pragati\Desktop\Uni X X + v

=====
1. Add Item
2. Display Inventory
3. Search Item
4. Update Quantity
5. Delete Item
6. Exit
Enter choice: 1

Enter ID: 1
Enter Name: soaps
Enter Quantity: 400
Enter Price: 50

? Item added successfully!

=====
INVENTORY MANAGEMENT SYSTEM
=====
1. Add Item
2. Display Inventory
3. Search Item
4. Update Quantity
5. Delete Item
6. Exit
Enter choice: 6

?? Exiting program...

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Process exited after 22.99 seconds with return value 0
Press any key to continue . . .
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CONCLUSION:

This project successfully demonstrates the design and implementation of an Inventory Management System using Linked List data structure. The system effectively addresses the core challenges of inventory management by providing a dynamic, efficient, and structured approach to storing and manipulating inventory records.

Through this project, the practical application of linked lists in solving real-world problems has been clearly established. The use of a linked list as the underlying data structure allowed for dynamic memory allocation, enabling inventory records to be inserted, deleted, searched, and displayed without the limitations of fixed-size arrays. Each node in the linked list efficiently stored all essential item details including Item ID, Name, Category, Quantity, Price, and Supplier Information.

The key operations implemented — insertion, deletion, searching, traversal, and display — proved that fundamental data structure concepts can be applied effectively to build functional and meaningful software solutions. The system successfully eliminates manual record-keeping errors, reduces data redundancy, and provides a fast and organized way to manage stock information.

From an academic perspective, this project strengthened the understanding of pointer manipulation, dynamic memory management, and algorithm design through hands-on implementation. It bridges the gap between theoretical data structure concepts and their practical, real-world applications.

In conclusion, the Inventory Management System using Linked List serves as a strong foundation for more advanced inventory solutions. With further enhancements such as file handling for data persistence, graphical user interfaces, database integration, and multi-user support, this system can be scaled into a fully functional enterprise-level inventory management application.

FUTURE WORK:

Although the current system fulfills its basic objectives, several enhancements can be made in the future:

- Database Integration — Replace the linked list with a database like MySQL to enable permanent data storage and faster retrieval.
- File Handling — Add file read/write functionality to save and load inventory data even after the program closes.
- Graphical User Interface (GUI) — Develop a user-friendly GUI using tools like Java Swing or Python Tkinter instead of a console-based interface.
- Search & Sort Enhancement — Implement advanced sorting algorithms like Merge Sort and search techniques like Binary Search for faster operations.
- Multi-user Support — Add login and role-based access control for Admin, Manager, and Staff users.
- Barcode/QR Code Integration — Enable barcode scanning for quick item lookup and stock updates.
- Report Generation — Add the ability to generate stock reports in PDF or Excel format.
- Cloud-based System — Host the system online to allow access from multiple locations in real time.